AI for Data and Data for AI

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GRAIL-V, A CVPR WORKSHOP

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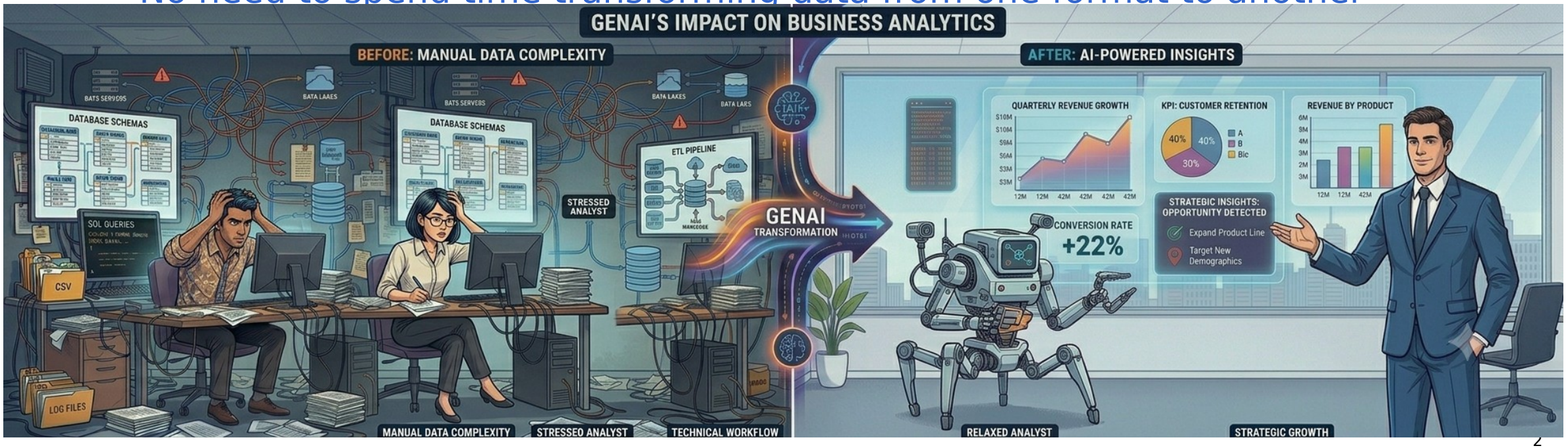
The Promise of Gen AI to Knowledge Access & Use

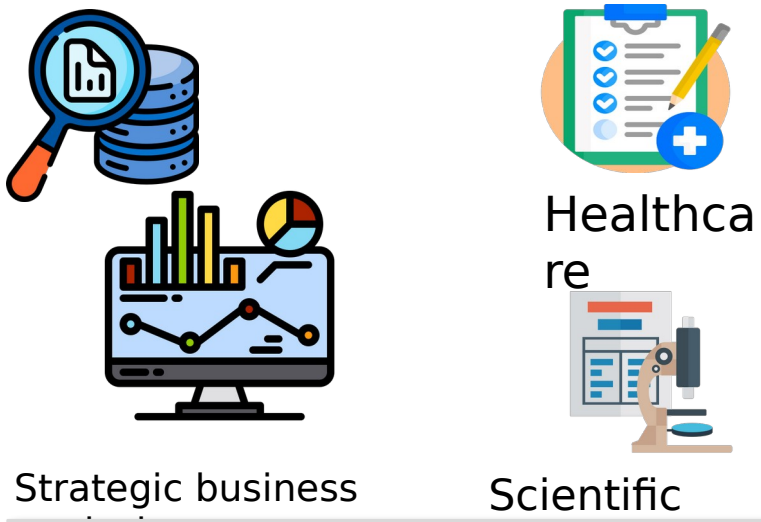
- GenAI promises to bring in a new wave of innovation in knowledge access & use.
- Users will solve their problems at the conceptual level – their terms & metrics
  - No need to learn domain specific query languages, database schemas
  - No need to spend time transforming data from one format to another



Knowledge Access & Use

- Models cannot do it alone
- **External Data** is essential for almost all ap

  - No more general-purpose data
    - The remaining data is owned by corporations, medical centers, people,…
    - It will stay **external** to the models



- The key use case for LLMs is going to be that of
  - Orchestrating and supporting **access and use of external data**
  - Multimodal data that models will not see* prior to run time.

[Coding] Agents need access to heterogenous knowledge sources

This Talk

commerce

Many Reasons:

Information Retrieval, Understanding Information Needs, Understanding the stored data, Using data.

- Reliably supporting decisions that depend

AI for Data and Data for AI

- **AI for Data**: Human Consumption
- **Data for AI**: AI Agents Consumption
  - Data needs to be organized & presented **differently**

- Instructions
- Structure and Format
  - Visualization provides information to humans; less to agents.
- Semantic Richness
  - Humans infer context from visual queues and commonsense
- Metadata and Context
- Granularity and Completeness
- Error Handling, Trust and Verification
- Data Preparation
  - Agents require a semantic layer, APIs with clear contracts

Why is it Hard?

- Core Difficulties:
  - Retrieval
    - The importance of a Semantic Layer
  - Dealing with Structure
    - NL-2-[SQL,…]

- Agentic Use
  - Multiple Knowledge Sources
  - Rich Documents
    - How to Present the Output?
  - Planning
  - Conflicting Information

- **Reasoning:** Underlying some of the difficulties are the needs to support specific types of reasoning that evade general purpose models

(1) Semantics Challenges to Re

The semantic layer is a key ingredient of realistic solutions for both structured and unstructured data. Even when we bring queries and data objects close together… "The Crowded Embedding Space.", ICML'26

- **Myth:** Dense Retrieval Works
- **Truth:** Retrieval systems do well only when th guage of the user query closely matches the target document
- Retrievers struggle to retrieve documents ose relevance to the query is implicit

wi ja

Are two eyes necessary for 3D vision? … Do people eventually lose the *depth perception* (or at least it diminishes significantly) when they lose a single eye?…

Human eyes are horizontally separated by about 50–75 mm (interpupillary distance) depending on each individual. Thus, each eye has a slightly different view of the world around. This can be easily seen when alternately closing one eye while looking at a vertical edge. The binocular disparity can be observed from apparent horizontal shift of the vertical edge between both views…

Retrieval using dense retrievers
→ The document ranks 67th

Semantic Layer: data objects have to be further **enriched** to facilitate better retrieval. LLMs can do it well, off-line: before

Chen et al. (COLM, 2025): **EnrichIndex:** Using LLMs to Enrich Retrieval Indices Offline

Many types of enrichments are needed – purpose, descriptions, QA, transaction-dependent, etc.

Find the highest last traded value of Alphabet in NYSE today

SECURITYID	ISTRADEON	ISPROVIDEDBY
1004374	2	2980
1002269	1	2374
…	…	…

Retrieval using dense retrievers
→ The table ranks 38th

[Shen et al. "LogiCoL: Logically-Informed Contrastive Learning for Set-based Dense Retrieval EMNLP'25]

(2) Structural Challenges: Table Retrieval

- Text 2 [SQL]

```sql
SELECT a.*
FROM accounts a
WHERE a.gender = 'Female'
  AND EXISTS (SELECT 1 FROM credit_cards cc WHERE cc.account_id = a.account_id)
  AND EXISTS (SELECT 1 FROM loans l WHERE l.account_id = a.account_id);
```

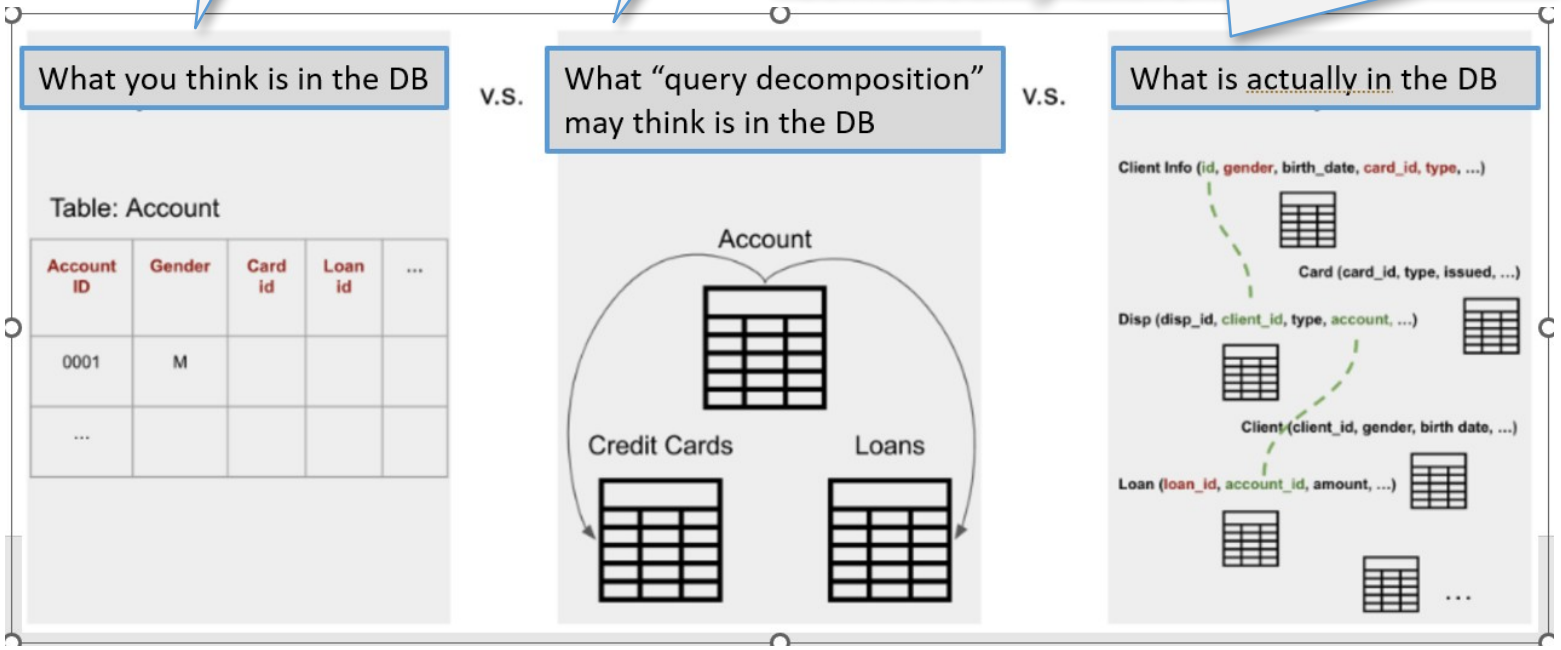
Who are the female account holders who own credit cards and also have loans?

Easy if there is a single table that is sufficient to answer the question

"Doable" if the query corresponds to how the information is stored

Understand the data organization process

Generally, the query may not align with how the information is organized across different tables; need to identify the join relationships while conducting table retrieval



Chen, Zhang, & Roth, Join-Aware Multi-Table Retrieval (ACL'24);

Chen et al. (ACL, 2025): Can we Retrieve Everything All at Once? ARM: An Alignment-Oriented LLM-based Retrieval Method

Information & Structural Alignment

What is the highest eligible free rate for K-12 students in the schools in the most populous county in California?

Unduplicated Student Poverty – Free or Reduced Price Meals Data 2023–2024

School Name	School Type	Educational Option Type	Enrollment (K-12)	Free Meal Count (K-12)	Percent (%) Eligible Free (K-12)
Envision Academy	K-12 Schools (Public	Traditional	223	154	69.1%
Community Schoo	Elementary Schools	Traditional	186	155	83.3%
Yu Ming Charter	Elementary Schools	Traditional	865	257	29.7%
Urban Montessori	Elementary Schools	Traditional	349	106	30.4%
Alameda County J	Juvenile Court Scho	Juvenile Court Scho	44	44	100.0%
Alameda County C	County Community	County Community S	58	40	69.0%

California Counties by Population (2024)

by Kristen Carney Updated:

The **most populous counties in California** are Los Angeles County with 9,663,345, San Diego Coun 3,269,973, Orange County with 3,135,755, Riverside County at 2,492,442, and San Bernardino Count 2,195,611. You can get the most current Census population data for all 58 California counties, sorted f **populous counties in California** to the least, according to the 2023 Population Estimates Program.

The table and document do not share overlapping information
 cannot be connected to answer the user question!

(2) Information & Structural Alignment

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Tables that are not explicitly mentioned in the user query but are essential l for connections!

California Department of Education
School and District Data Directory Export
Educational Data Management Division
December 9, 2024

Record Type	CDS Code	School	Funding Type
District	01407900000000	No Data	No Data
District	01100170000000	No Data	No Data
School	01100170130419	Alameda County Community	No Data
School	01100170130401	Alameda County Juvenile Hall/Court	No Data
School	01100170130427	Alameda County Opportunity	No Data

Fiscal Year 2024–25
California Department of Education
Legend: County District School (CDS) code; N/A = Not a component of another district because district [*EC* Section 35515]

CDS Code	County	District
01611190000000	Alameda	Alameda Unified
01611270000000	Alameda	Albany City Unified
01611430000000	Alameda	Berkeley Unified
01611500000000	Alameda	Castro Valley Unified
01750930000000	Alameda	Dublin Unified
[illegible]	Alameda	Emery Unified

Involves hypothesizing a KG at run time

(2) Information & Structural Alignment

The top two data objects can be retrieved by **semantic similarity** to the decomposed parts of the query.

The bottom two table cannot. Retrieving them requires an iterative process that considers **joinability**

students in the schools in the most populous county in California?

"Standard" RAG isn't sufficient – does not consider internal da

Agentic RAG (ReACT) can get stuck, for the same reason.

Optimal retrieval system

o n alignment

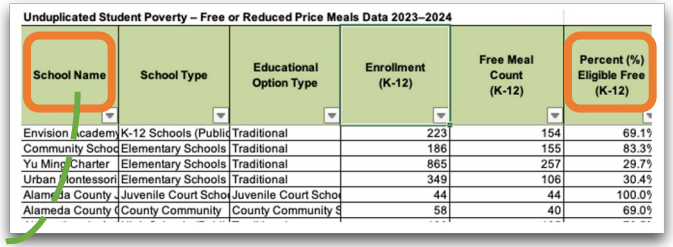
Structur al alignme nt

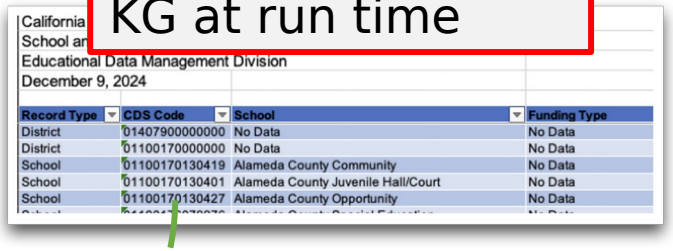
Involves hypothesizing a KG at run time

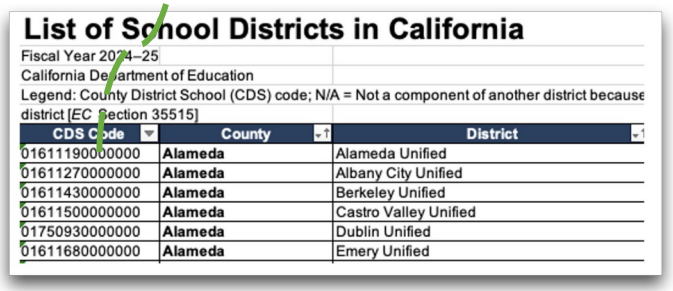
This is a form of Enrichment, but some of it needs to be done at run-time, given a query. (But we can cache it)

More can be done – e.g., with the use of **query logs**

Peter Chen et al. : Log-augmented generation: Scaling Test-Time Reasoning with Reusable Computation (ICLR'26)





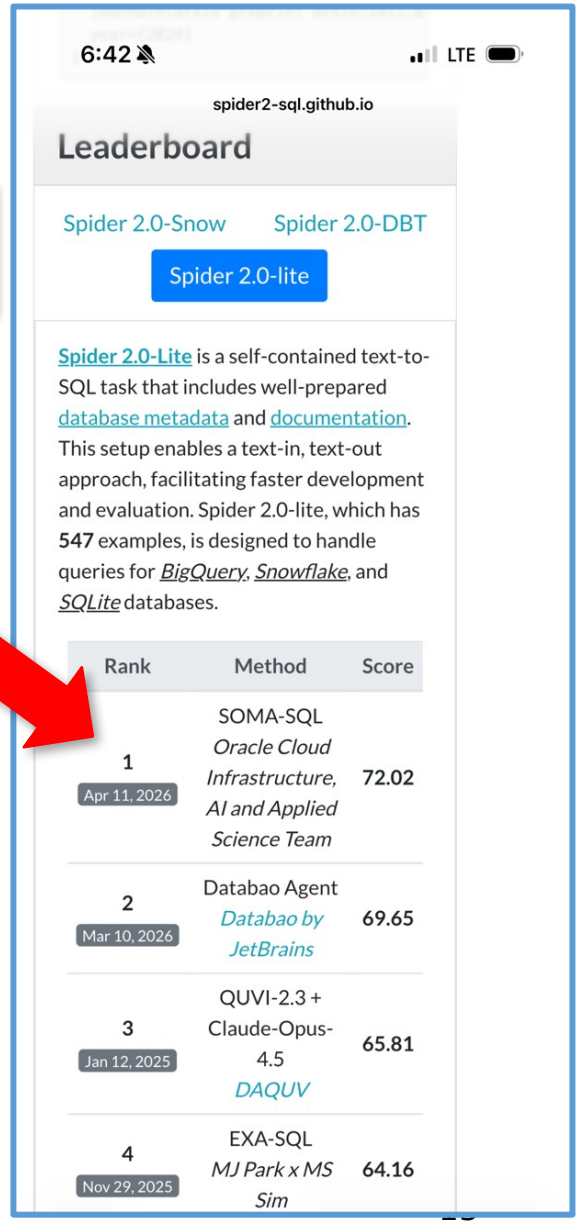


Where are we in NL-2-SQL?

- **ARCHER:** recently introduced Text-to-SQL benchmark with complex arithmetic ar tasks
  - 9/19/25: A vanilla GPT-5 based baseline: 44.23%
    Oracle AI Agentic system: 54.96%
    [NEWS] OCI AI Science wins first place in the 2025 competition!

Look at the absolute results!

- A vanilla NL2SQL system: [NL Query LLM SQL] is not sufficient [in most cases].
  - Performance is dataset specific; failures are due to both retrieval and generation
  - Gaps between vanilla models and advanced systems ~20-50%



Why is it Hard?

Many exciting problems to think about

- In order to access the knowledge we need, and do it efficiently, it is essential to
  - (1) Develop an **abstract semantic layer** on top of existing data sources (to bring the data closer to the needs), and
  - (2) **Structure:** Understand **how the data is organized** so that we can plan well

- On the Query side:
  - **Semantic similarity:**
    - Queries and data objects are not "similar", and even when they are often it's not sufficient.
  - **Complex queries:**
    - Require decomposition that is often structure aware

- On the Data (base) side:
  - Schemas could have a **complex structure**
    - Difficult to retrieve the right tables.
  - **Naming:**
    - Often ambiguous and uninformative
    - Entity linking

- **Reasoning:** Queries could require supporting specific types of reasoning that evade general purpose models
  - Aggregation of information, temporal references and reasoning, counterfactuals,...

ORACLE AI Solution: SOMA-SQL

[Somayajula, Liu, et al.'26]

Synthetic Query Log and Probing for Multi-source Ambiguity Resolution

- NL Question: "What were the top-performing regions in New York last quarter?"
  - Query-level ambiguity – underspecified intent:
    - **New York:** city, state, NYSE value
  - Schema-level ambiguity – unspecified reference:
    - **Last quarter:** calendar, fiscal, rolling 90 days
  - Contextual ambiguity – questions assu business logic stion and the s

Different choices impact filters, joins, and time windows.

- **SOMA-SQL Workflow:**
- Detect ambiguity dimensions
  - Taxonomy cues + K-SQL disagreements
- Resolve with evidence
  - Synthetic-log MCQs + database probes
- Generate evidence-backed SQL

Net effect: an ambiguous question becomes an evidence-backed intent, not a silent guess.

Differentiator #1: Synthetic Query Logs

- SOMA-SQL converts historical SQL into MCQ-style disambiguation tool

OFFLINE: Construct Synthetic Query Log

Validated SQL log (intent anchor) → **Detailed NL intent** (fully specified NLQ) → **Ambiguous NL wording** (remove details) → **K SQL variants** (interpretation samples) → **MCQ record** (stored resolution)

Validated SQL anchor

```
SELECT r.region_name,
  SUM(o.revenue) AS total_revenue
FROM orders o ...
WHERE c.state = 'New York'
  AND fiscal_q = '2024 Q1'
```

→ **Detailed NL intent**
List top-5 regions in New York State ranked by total revenue in fiscal quarter Q1'24

→ **Ambiguous user wording**
Find top-5 performing regions in New York last quarter?

→ **Candidate variants**
A SUM(o.revenue)
B SUM(o.profit)
C COUNT(o.deal_id)

→ **Stored MCQ record**
How should "performing" be ranked?
A) Revenue ✓
B) Profit
C) Deal count

ONLINE: Retrieve MCQs to Resolve a New User Question

New user question
"Can you help me find the top-performing region last quarter?"

→ **Retrieve relevant MCQ**
How should "performing" be ranked? → Revenue

→ **Use as generation context**
Inject clarification into planning and SQL generation; produce intent-conditioned SQL

Key takeaway: historical SQL expressions provide useful clarification knowledge for disambiguation

Differentiator #2: Ambiguity-Driven Probing

- SOMA-SQL probes plausible interpretations against DB & uses it to repair the final SQL.

ONLINE: Resolve Ambiguity with Database Evidence

Ambiguous question
"show me churn for enterprise customers last quarter"

→ **Generate** Competing probe SQLs

→ **Execute & Resolve** With probe report

→ **Repair** Final SQL

Ambiguity: What does "churn" mean in this database?

① Canceled subscriptions

```
SELECT COUNT(DISTINCT
s.customer_id)
FROM subscriptions s JOIN
customers c USING(customer_id)
WHERE c.segment='Enterprise'
  AND s.status='canceled'
  AND s.cancel_date >= :q_start
  AND s.cancel_date <  :q_end;
```

→ **4,821**

② Inactive accounts

```
SELECT COUNT(DISTINCT
a.customer_id)
FROM accounts a JOIN customers
c USING(customer_id)
WHERE c.segment='Enterprise'
  AND a.last_activity_date
< :q_start
  AND a.account_status='active';
```

→ **137**

③ No recent purchase

```
WITH last_order AS (
 SELECT customer_id,
MAX(order_date) last_dt
 FROM orders GROUP BY 1)
SELECT COUNT(*) FROM customers
c LEFT JOIN last_order lo
USING(customer_id)
WHERE c.segment='Enterprise' AND
lo.last_dt < :q_start;
```

→ **94**

✓ **Probe report selects strongest evidence**
Canceled subscriptions has the strongest schema-supported

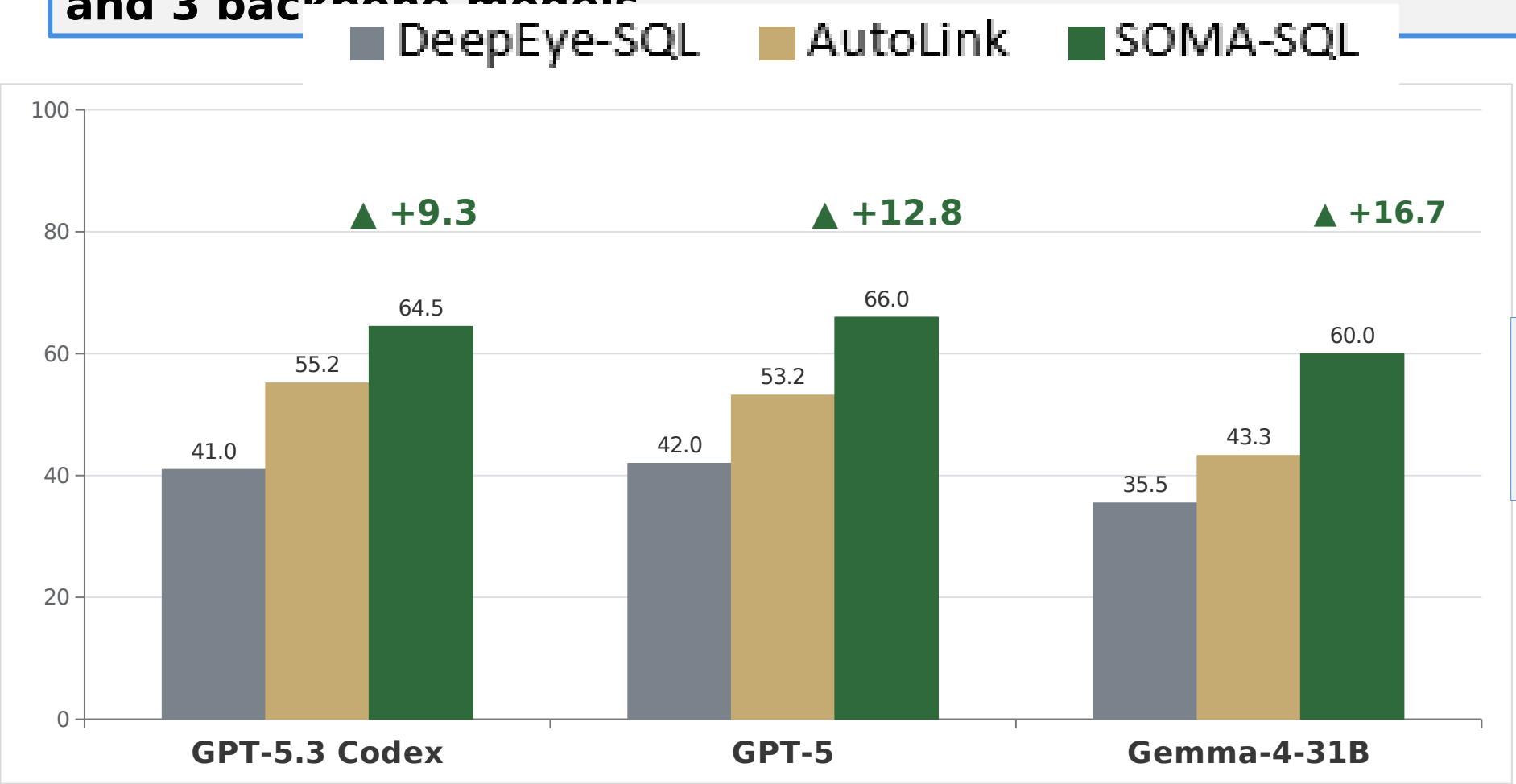
→ SQL

Final SQL uses resolved intent
Apply cancellation filters; remaining probes remain audit evidence.

Key takeaway: ambiguous terms are resolved from database evidence, not guessed, combined for final SQL.

Consistent Gains Across Diverse NL2SQL Benchmarks

Average EX across 5 benchmarks, 4 SQL dialects, 2 languages, and 3 backbone models



+9.3 to +16.7
Execution Accuracy gain over strongest baseline

Values are execution accuracy (EX%). Gains shown above SOMA-SQL bars are relative to the second-best method.

Benchmarks: Spider 2.0 Lite, BIRD, Archer-en, Archer-zh, BEAVER

Why is it Hard?

- Core Difficulties:
  - Retrieval
    - The importance of a Semantic Layer
  - Dealing with Structure
    - NL-2-[SQL,…]

- Agentic Use
  - Multiple Knowledge Sources
  - Rich Documents
    - How to Present the Output?
  - Planning
  - Conflicting Information

- **Reasoning:** Underlying some of the difficulties are the needs to support specific types of reasoning that evade general purpose models

Using Multiple Information Sources

- (More) Natural and Complex question answering
  - Decomposition into intermediate questions & Planning
  - Retrieval from heterogeneous information sources: Text, Tables, Lists, Images
  - Aggregation and Reasoning over intermediate answers.

Q: In EU countries, do right-wing parties tend to favor women leaders more than left-wing parties?

the political left than on the right in EU countries. There are 46 women leaders of political parties in EU member states [a]. Of those 28 belong to left-wing and center-left political parties [b].

Retrieval

Multi-document evidence:

[a] The list of female party leaders in EU countries includes: Li Andersson [a.1], Giorgia Meloni [a.2], Lilian Marijnissen [a.3], …
[b] The women-lead EU parties which belong to the political left are: Left Alliance [b.1], Green League [b.2], …

[a.1] Li Andersson: "The leader of the Left Alliance, she has been a Member of Parliament since…."
[a.2] Giorgia Meloni: "... she became president of FdI…"
[a.3] Lilian Marijnissen: "... serving as Leader of the Socialist Party…"
…

[b.1] 2023 Finnish election results:

Party	Votes	%	+/–	Seats
National Coalition Party	644,555	20.82	+3.82	48
Finns Party	620,981	20.06	+2.58	46
Social Democratic Party	617,552	19.95	+2.22	43
Centre Party	349,640	11.29	–2.47	23
Green League	217,795	7.04	–4.45	13
Left Alliance	218,430	7.06	–1.11	11
Swedish People's Party	133,518	4.31	–0.22	9

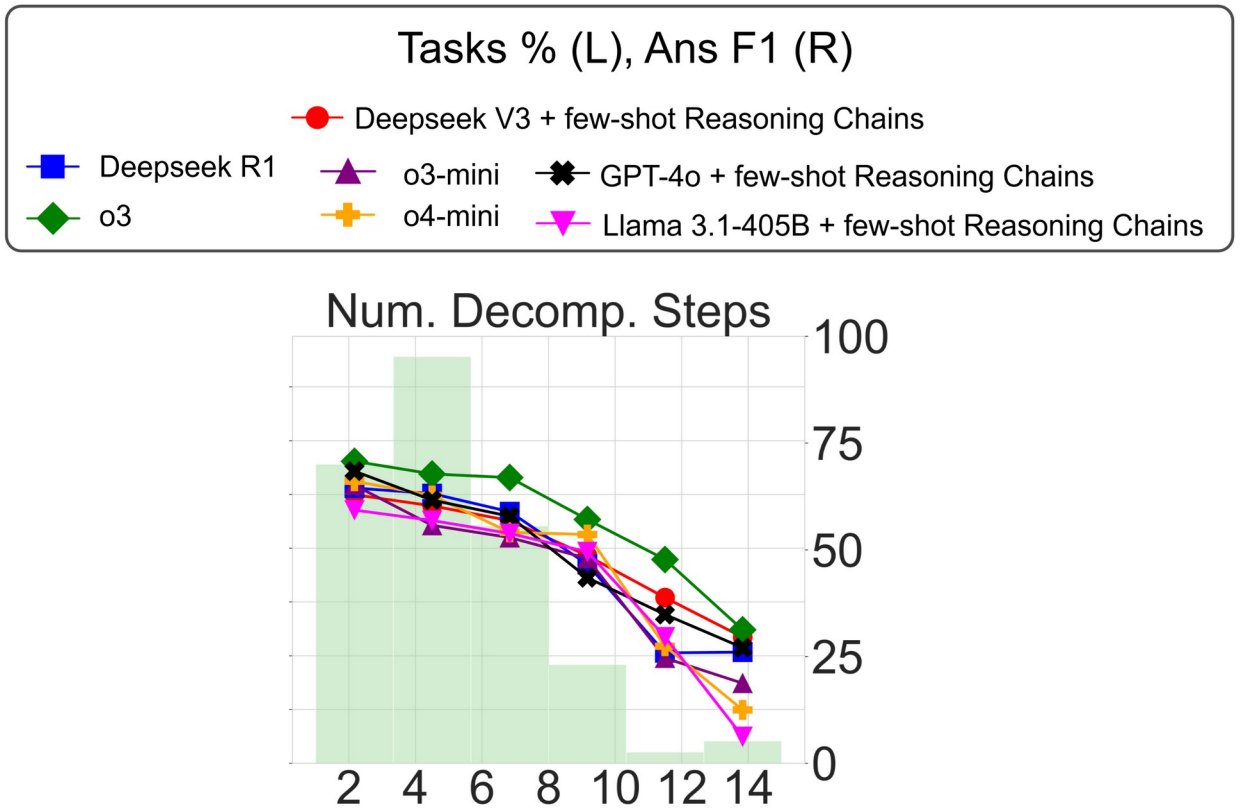
[b.1] Left Alliance (Finland): The Left Alliance is a member of the Nordic Green Left Alliance and an observer member of the Party of the European Left.
…

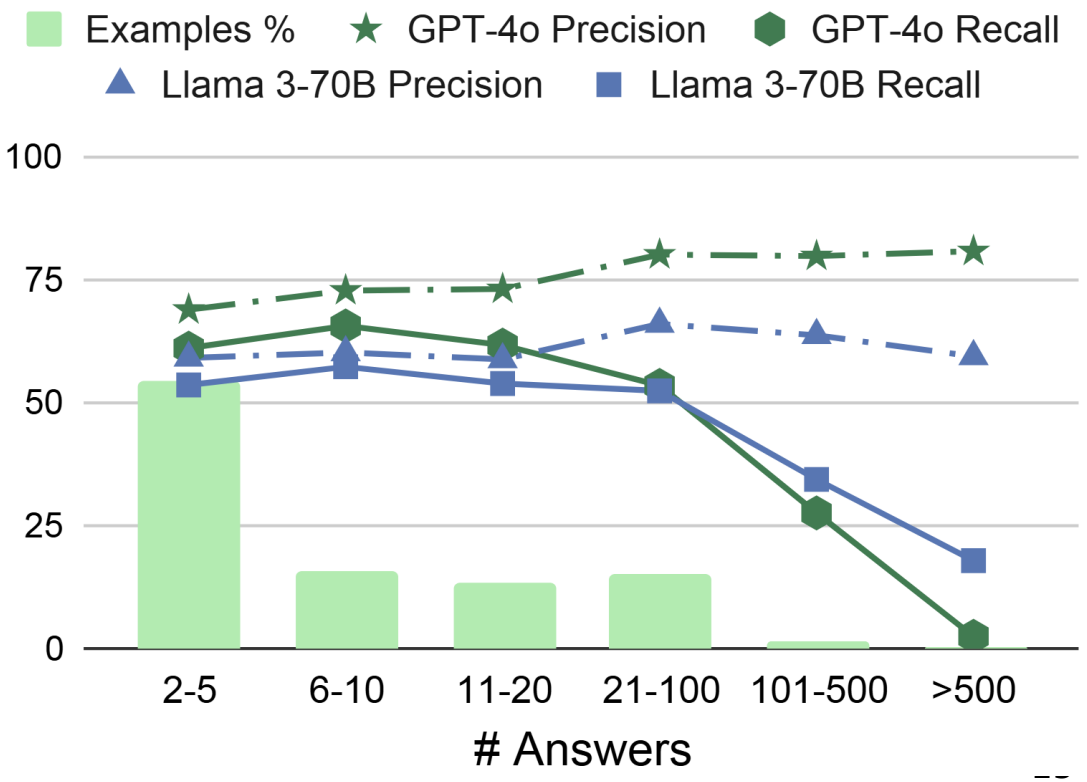
Tomer Wolfson et al.: The MoNaCo Benchmark (TACL, 2025)

Models Struggle on Multi-Step Reasoning

- Frontier LLMs suffer as a function of:

more reasoning steps (left) as well as **more**





A Realistic Electronic Health Records Example

Show me all my patients whose seizure frequency increased by more than 50% over the past 3 months, list their current medications, flag any that are non-adherent, and summarize what w

Show me all my patients whose seizure frequency increased by more than 50% over the past 3 months, list their current medications, flag any that are non-adherent, and summarize what we discussed in their last visit.

entl ist)

Perhaps the key challenge to agentic approaches today

But it all depends on retrieving the information needed

Summarize last visit
Function: getLastVisitSummary(patientList)

The Rich Data Challenge

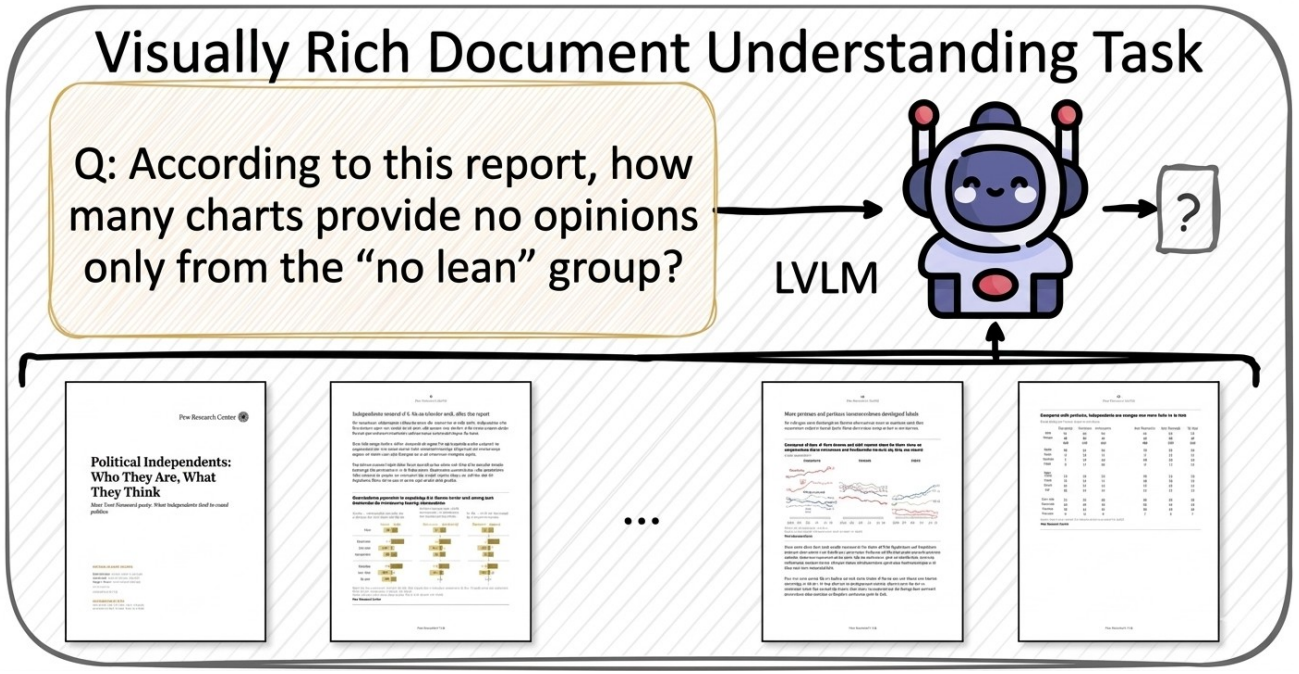
- In many cases relevant information is not exposed during retrieval
  - Too costly to do everything at inference time
- Ingestion becomes a bottleneck – a form of entrichment
  - Pre-process documents and enrich with (expected) relevant knowledge

[Sourati et al. ACL'26]: Visually Rich Documents

- Retrieval over structured documents requires both semantic understanding and explicit structural reasoning.
  - Layout, section hierarchy, element types, and cross-page links carry meaning, and must be captured at ingestion time.
  - Once structure is discarded at ingestion, no retrieval strategy can recover it.
- Build an explicit, holistic document representation during ingestion
  - Exposes to retrieval information & structure that otherwise would not be accessible .

Visually Rich Document Understanding

- Documents combine text, figures, tables, and charts
- Meaning depends on layout and reading order
  - A caption only makes sense next to its figure
  - A table cell only makes sense under its header

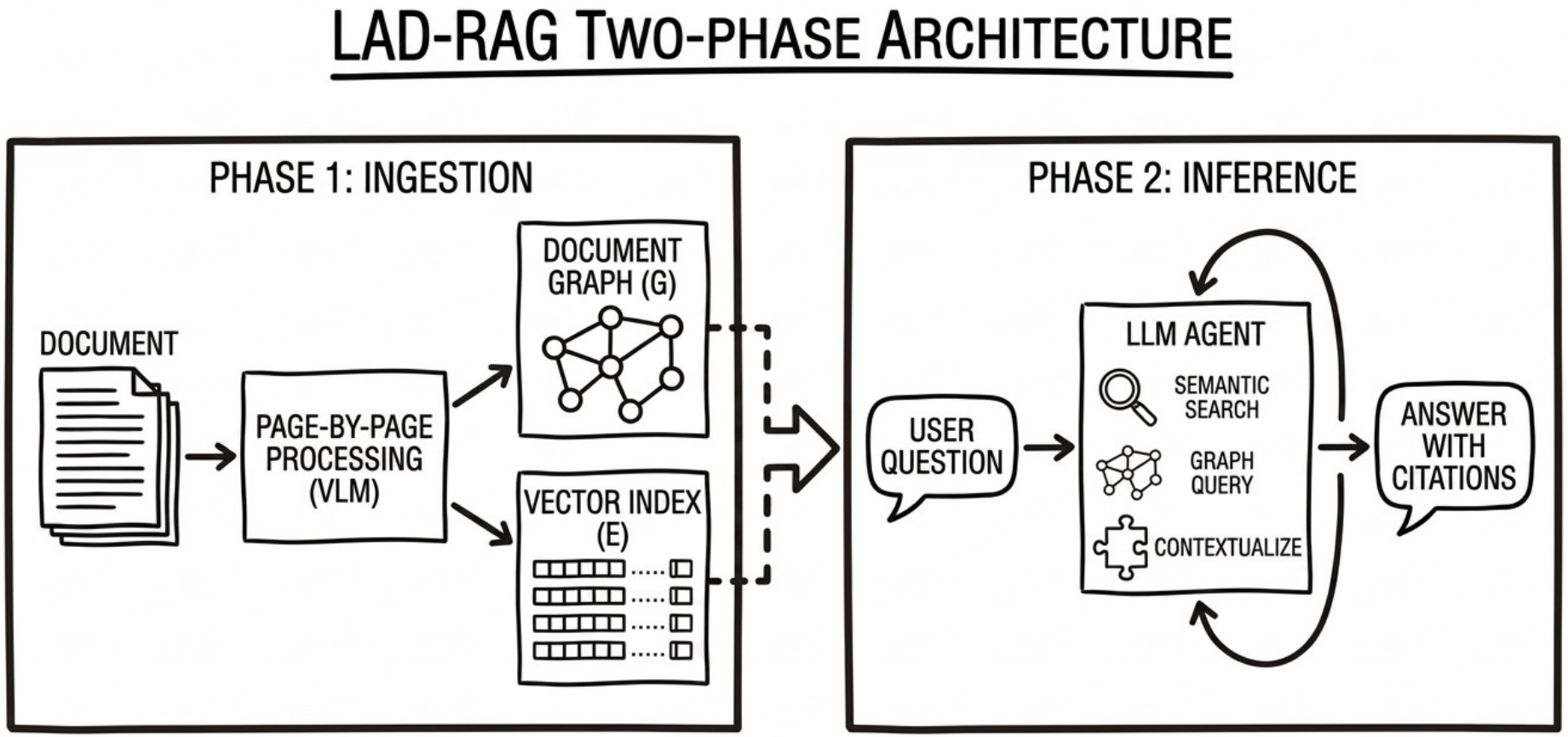


Off-Line & Inference-Time Processing

- Ingestion:
- VLM processing page by page.
- For each page, extract visible elements:
  - paragraphs, tables, figures, section headers, footnotes,
- Generate rich metadata for each:
  - Positions; types; content; summary
- Generate two indices:
  - Symbolic index: structured properties and explicit relationships.
  - Neural index: Vector embeddings of the self-contained summaries of all nodes.

- Inference:
- Uses an LLM agent to iteratively search across both indices, with three tools:
- NeuroSemanticSearch(query):
  - Standard embedding similarity search.
- SymbolicGraphQuery:
  - Structured queries over the document graph.
- Contextualize:
  - Given a node, expand to related content via graph structure. (Louvain community detection)

Off-Line & Inference-Time Processing



LAD-RAG consistently outperforms all retrieval baselines across multiple benchmarks (multiple models) both at Retrieval evaluation and end-2-end QA performance.

Planning & Optimization

- There is a lot to say about the computational issues underlying planning
  - Constraint Optimization Problem: Models cannot do it well without translation to external solvers [Wang et al.'26]

- Planning for accessing, aggregating & organizing data for AI
  - The challenge: Planning to access multiple knowledge sources is typically done with incomplete information

And How About Coding Agents?

Give me a list of the European male Tennis Players sorted by their tennis income in 2025

I now have solid data across multiple sources. Here is the ranked list of **European male tennis players by 2025 prize money:**

Rank	Player	Country	2025 Prize Money
1	**Jannik Sinner**	IT Italy	$19,120,641
2	**Alexander Zverev**	DE Germany	$6,060,174
3	**Novak Djokovic**	RS Serbia	$5,140,175
4	**Lorenzo Musetti**	IT Italy	~$4,682,629
5	**Jack Draper**	GB Great Britain	~$3,421,704
6	**Casper Ruud**	NO Norway	~$3,268,588
7	**Andrey Rublev**	RU Russia	~$2,877,831
8	**Holger Rune**	DK Denmark	~$2,973,472
9	**Daniil Medvedev**	RU Russia	~$2,877,831
10	**Karen Khachanov**	RU Russia	~$2,774,481
11	**Jakub Menšík**	CZ Czech Republic	~$2,652,860
12	**Flavio Cobolli**	IT Italy	~$2,623,372
13	**Jiří Lehečka**	CZ Czech Republic	~$2,596,590
14	**Alejandro Davidovich Fokina**	Spain	~$2,692,725

- One key difficulty is **planning:** some plans are easier to execute than others.
  - Should we start with a list of tournaments or players?
- Depends on understanding the data in the domain

Anyone missing?

Which is bigger?

Identical Income?

- **Coding Agents are doing things faster than we can**
- **But they are prone to errors & high cost**
  - From **safety** to **mistakes** – especially when the tasks involve knowledge

Conflicts, Perspectives, and Mis-/Disinformation

"*What is the occupation of George Washington?*"

A1: George Washington (February 22, 1732 - December 14, 1799) **was an American Founding Father, military officer, and politician** who served as the first president of the United States from 1789 to 1797
A2: George Washington (born October 18, 1907) was an **American jazz trombonist.**

"*Had to remind him to toast the sandwich*"

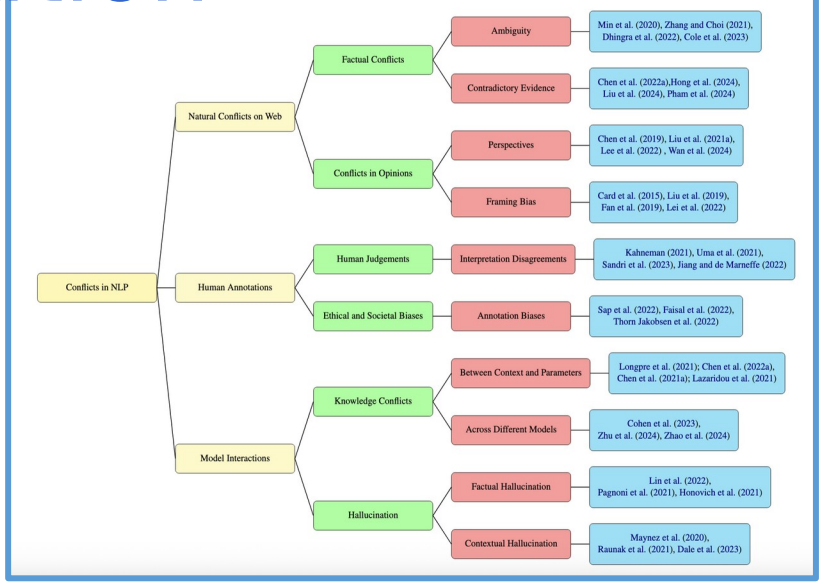
Majority Label: **Negative**
Minority Label: **Neutral**

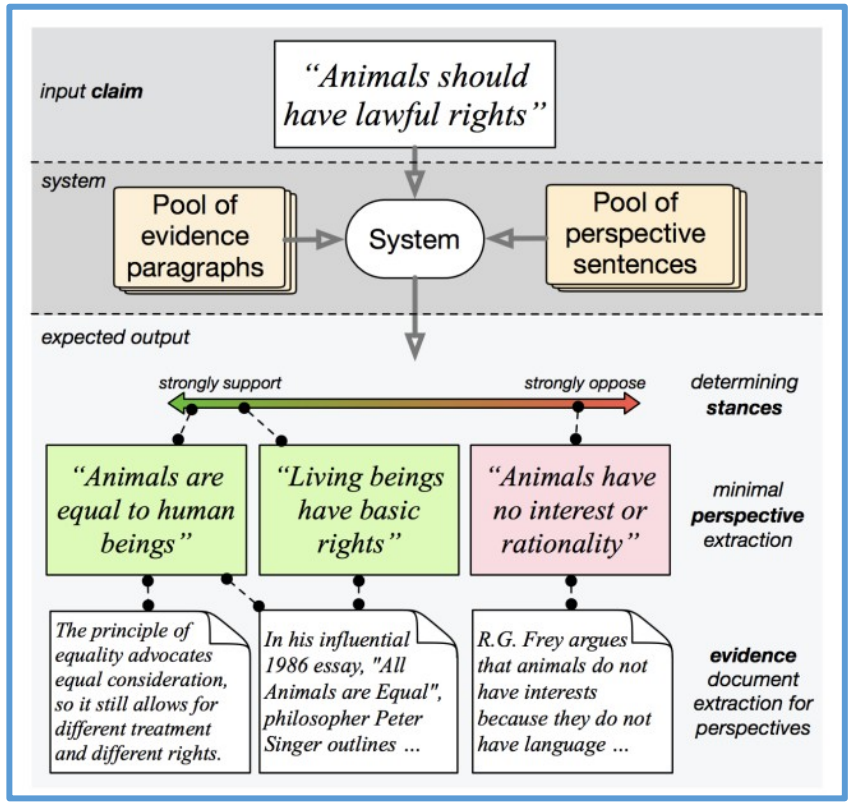
"*Who is the current CEO of Amazon Web Services (AWS)?*"

Context: **Matt Garman** is the CEO of Amazon Web Services (AWS), starting June 2024
Memory: **Andy Jassy** is the CEO of AWS.

Figure 1: Examples of the three different ares of conflicts discussed in this work. The first example describes a case where two different entities of the same name are found *naturally on the web*, the second example elaborates the *annotation disagreement* in a sentiment analysis task, and the third showcases a knowledge conflict between the context and memory of LLMs during *model interactions*.

Conflicts in Texts: Data, Implications and Challenges, Liu & Roth, EMNLP 2025





Sihao Chen, et al. **Seeing things from a different angle: discovering diverse perspectives about claims**.

Siyi Liu et al., **Open domain question answering with conflicting contexts**. NAACL 2025

Conclusion

Thank You!

Real-world data is often **multimodal**, **temporal**, **multilingual**, and it is still challenging to access, understand, and use it reliably

- The key use case for LLMs is going to be that of
  - Orchestrating and supporting **access and use of external data**
  - Multiple heterogenous knowledge sources that models will not see prior to run time

And the consumers are going to be both humans and AI agents.

There is work to do in core problems – from retrieval, to structure, to planning and dealing with conflicting information.

With Vivek Gupta, et al.